Exploring Contingency: Retrieving Disinformation Narratives with Speculative Fiction Generation

Anonymous ACL submission

Abstract

As generative system research shifts toward integrating interpretation techniques, narratives are gaining prominence, especially in disinformation and propaganda detection. While several datasets targeting narrative classification on topics such as Russia's war on Ukraine, climate change denial and obstruction were released in the last years, the task of retrieving disinformation texts by narrative is largely understudied. In the following, we propose repurposing narrative classification datasets for the evaluation of retrieval strategies. We analyze the datasets using metrics such as readability and narrative distinctness. We also introduce two new metrics in this study: narrative contingency and narrative variance. We discuss performance metrics and propose a system inspired by speculation-based techniques, similar to Hypothetical Document Embeddings (HyDE). Furthermore, we consider the addition of graph-based reasoning approaches within dynamic few-shot sampling and how models with safety alignment removed in post-training perform differently in this setup, resulting in a relative gain of 62.57% on the MAP metric of our best performing setup over dense retrieval. Finally, we explore the statistical correlation between retrieval performance, readability, and narrative metrics to advance the development and refinement of narrative taxonomies.

1 Introduction

A recent consensus paper published by a consortium of 38 senior researchers (Kommers et al., 2025) argues for a paradigm shift within the research on generative systems towards hermeneutics: the science of interpretation, a central approach in the humanities. The emerging field of computational hermeneutics calls for an advancement of understanding within transformer-based systems by accounting for the process of meaning-making being highly dependent on sociocultural context and the possibility of the existence of multiple, sometimes conflicting interpretations of the same source. Within the realm of information retrieval, this paradigm shift can be envisioned as a move towards the enhancement of meaning-based retrieval by incorporating interpretation techniques. Although semantic search with sentence embeddings does enable to retrieve meaning beyond lexical features, it still remains superficial. In general, current language models have difficulties with interpreting subtext. For example, attempts to capture authors' intent in literary summarization tasks have shown mixed performance (Subbiah et al., 2024). Embedding models also struggle with information on a subtext level, such as implicit logical constraints formulated in queries (Shen et al., 2025), or the understanding of relations between concepts (Zhang et al., 2023).

In the following, we explore the notion of retrieving texts by their implicit meaning within the domains of disinformation and propaganda detection. The notion of "narratives" gained traction in recent years within these fields and was studied in works such as Solopova et al. (2023b,a), where different narratives and their linguistic features were considered in the context of automated propaganda detection. Such works represent an increased effort of identifying common patterns across disinformation and propaganda texts and are accompanied by studies such as Lei and Huang (2023), which indicate that propaganda narratives can also follow certain common discourse structures or Holur et al. (2022), where narrative features of conspiracy theories such as common roles, characters, and their relations are highlighted.

At the same time, several datasets such as Coan et al. (2021), Rowlands et al. (2024) or Nikolaidis et al. (2025) emerged aiming at the idea of classifying disinformation texts by narrative. In this study, we explore how these datasets can be repurposed for evaluating narrative retrieval strategies. We propose a new point of view in regard to these datasets to enable use cases such as free-text monitoring of narrative development in large text corpora over time, by using narrative taxonomy labels as queries and associated text instances as potential retrieval candidates. This allows us to evaluate specific narrative retrieval approaches on already existing human-annotated data. We opt for the term *narrative retrieval* to emphasize the interpretative nature of this task, which within this study focuses on assessing key messages behind a text that might not be tangible on a lexical and a surface semantic level.

We explore this idea through a retrieval system we designed, drawing on insights from the interpretative sciences. Text interpretation is non-trivial for both machines and humans, forming a method and an object of study in narratology, hermeneutics, semiotics, and discourse and literature analysis. Within the latter, one framework for interpreting texts is based on speculation grounded in formulated premises (Roine, 2020). We find an equivalent of this idea in the retrieval strategy proposed in Gao et al. (2023), where hypothetical documents are generated to improve retrieval results. Another key component of our system draws on Hellman (2024), framing narratives as socio-cultural systems of interconnected stories with features like actors, events, and locations, which can be modeled as a graph. We follow the intuition that reasoning on such data could improve retrieval results and explore a specific feature of the graph-based retrieval framework NodeRAG (Xu et al., 2025), where high-level summaries of detected node communities are generated. In our system, we re-use this summaries as few-shot samples for generating hypothetical documents. The main goals of this study are to advance the understanding of "narratives" within disinformation detection and to understand how they are used to identify disinformation patterns. Thus, the contributions of this work are aligned with answering the following research questions.

- RQ1: How can datasets developed for narrative classification be re-purposed for evaluating narrative retrieval?
- RQ2: Can narrative retrieval be improved by speculation-based interpretation techniques?
- RQ3: Is there a correlation between retrieval performance, readability metrics, and embedding-based narrative metrics?
- RQ4: Do LLMs with removed safety alignment lead to better performance when generating hypothetical disinformation documents?

2 Preliminaries and Related Work

2.1 Disinformation Narratives Classification and Retrieval

In practice, the term "narrative retrieval" is used mainly to mean claim retrieval, focusing on individual claims, not overarching key messages, such as in Singh et al. (2024); Singh (2024). Akter and Santu (2024) identified the need to develop metrics for evaluating similarities between narratives in retrieval settings, but at that time were lacking a large enough dataset with narrative pairs to evaluate their hypothesis comprehensively.

Recently, several datasets considering narratives in disinformation such as (Heinrich et al., 2024) or Haouari et al. (2025) were released. The strategies for constructing these datasets share a common focus on grouping texts by narrative, often organized across multiple levels of abstraction. The structure is inherently hierarchical, following a many-to-one relationship, where individual texts are aggregated into progressively larger and more abstract, predefined groupings (e.g. (1) topics, (2) broad narratives, (3) narratives, such as in Sosnowski et al. (2024)). Albeit similar in the general strategy, there is no consistent labeling scheme across these datasets. In CARDS (Coan et al., 2021), the 2 levels of abstraction are defined as *claims* and *superclaims*. In Kotseva et al. (2023), a similar concept is referred to by a three-level hierarchy of *supernarrative*, *narrative* and *subnarrative* terms. While these datasets were developed for the classification of texts by narrative, we aim at re-using them to dynamically retrieve members of those groups by using their key messages as queries within the following study.

2.2 Semantic Search: Retrieving by Meaning

Due to its prominent role in Retrieval Augmented Retrieval (RAG) pipelines (Gupta et al., 2024), semantic search is currently at the center of significant research interest, with numerous approaches being actively explored to enhance its performance. The optimization of key factors like chunk sizes or the construction strategy of knowledge bases (Li et al., 2025a) is actively discussed. While dense retrieval is commonly applied within RAG systems

(Yu et al., 2025), it can still be outperformed by sparse retrieval methods such as BM25, as demonstrated by Weller et al. (2025) or Ren et al. (2023). In fact, studies report that in tasks that depend more on understanding the meaning of texts, such as argument retrieval, the superiority of BM25 was also observed (Thakur et al., 2024). Another direction of current RAG research focuses on adapting reasoning techniques within retrieval (Li et al., 2025b), motivated by the necessity of a deeper understanding of queries and potentially relevant documents. Working within a similar research gap, datasets for reasoning-intensive retrieval tasks such as BRIGHT (Su et al., 2025) were published. The idea of narrative retrieval explores a similar direction, highlighting the need for more complex reasoning akin to human interpretation techniques.

2.3 Generating Hypothetical Documents for Retrieval Tasks

One strategy of interpreting narratives from a humanities perspective is the application of speculation. In Roine (2020), narratives and speculation are introduced as complementary, where speculation aims at engaging with what is possible. In the proposed framework, the (1) *premise* (our in our context: key message) serves as a starting point for the exploration of possibilities. It serves as the base for (2) *thought experiments*, the actual work of fiction and the formulation of possibilities as consequences derived from the premise. Which are followed up by the creation of the concrete (3) *media* that gives form to the content.

This "instrumental mode of speculation"(Roine, 2020) has an equivalent within the information retrieval domain: the retrieval method Hypothetical Document Embeddings (HyDE) introduced in Gao et al. (2023), which relies on generating possible documents based on an input query and thus can be understood as an incorporation of the aforementioned mode of speculation. HyDE follows the strategy of generating n hypothetical documents based on a user query, retrieving n corresponding results, and merging these to return a final top-k results, effectively accumulating concepts and ideas related but not present in a query. Compared to Roine's framework, the query functions as the premise, while the n hypothetical documents correspond to the thought experiments. In our case, the medium is represented by texts found in social media and news media containing disinformation.

2.4 Graph-based Reasoning in Retrieval

Hellman (2024) understands narratives in the context of disinformation as interconnected systems of stories which involve features like actors, events, plot elements, and other narrative features. This notion is similar to Piper et al. (2021), where the work on such features is surveyed from a computational narratology perspective. The understanding of narratives as systems can be translated to sets of graph data within computational domains. In RAG, several graph-based approaches were recently proposed to enhance retrieval pipelines with extracted and graph-structured knowledge (Li et al., 2025b). Here, one example of seminal work, GraphRAG (Edge et al., 2025), prompted for subsequent approaches such as PathRAG (Chen et al., 2025), LightRAG (Guo et al., 2025) or NodeRAG (Xu et al., 2025), which optimized the high computational costs of GraphRAG and refined other aspects, like the strategy of graph construction. Within our proposed system, we are mainly interested in a specific feature from the NodeRAG framework. Similar to GraphRAG, node communities are detected the constructed knowledge graph, grouping graph data to generate community summaries. In GraphRAG, these summaries provide abstraction during the retrieval process and serve as coarse and hierarchical semantic layers between a query and graph structures possibly related to the query. In NodeRAG, where a heterogeneous graph is constructed to enable graph propagation methods along different types of nodes, high-level community summaries are generated on heterogeneous nodes to become a part of the graph as nodes to be retrieved.

3 Methodology

3.1 Proposed Architecture

The system we propose to address our research questions is illustrated in Figure 1. The use case scenario modeled towards narrative monitoring incorporates two data sources: A preexisting knowledge base, which we simulate with the training splits of the datasets we use, and an incoming data batch: the data corpus we want to retrieve narratives from, simulated by the test splits. Our goal is to retrieve top k results based on a narrative description constructed from the dataset label taxonomies as a query (the query construction steps per dataset can be found in the Appendix):

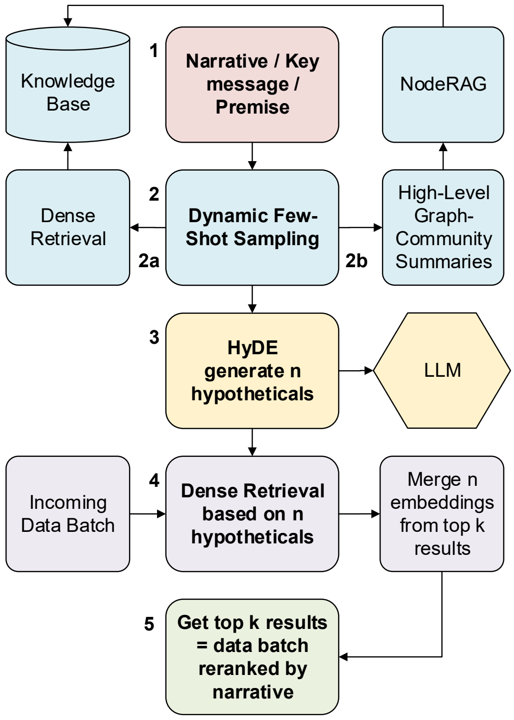

Figure 1: Architecture of the proposed system

1. Use a “narrative” from a dataset as an input
2. Dynamically collect example pairings between narratives and texts from the knowledge base, without relying on labels
   a) Dense Retrieval: Get top 1 candidates per narrative by cosine similarity
   b) NodeRAG: Get top 1 high-level community summaries
3. Use the retrieved examples as few shots in generation prompt of hypothetical documents; generate $n = 10$ documents
4. Retrieve top-k results based on n hypothetical documents with dense retrieval and average the results across embeddings
5. Return the merged top-k results

For step (2), we propose a variation of few-shot sampling strategies, where examples are collected in prompts to guide the generation of LLMs. Since we aim to build our system as dynamic as possible to advance towards free-text narrative monitoring, we sample potential examples at runtime (similarly to Ge et al. (2025)) based on only the texts from the knowledge base (the training set) without assessing the labels. The examples are retrieved either by (2a) running dense retrieval on the knowledge base, or by (2b) retrieving high-level graph-community summaries from NodeRAG. For our experiments, we opted to work with NodeRAG over GraphRAG because of the higher nuance of the community descriptions due to more steps of reasoning with heterogeneous graph data and due to NodeRAG outperforming GraphRAG in retrieval metrics while requiring a significantly smaller amount of tokens (Xu et al., 2025). Once the few-shot samples are collected, the following steps (3), (4) and (5) are aligned with the original HyDE strategy as proposed in Gao et al. (2023) with the additional change of following Zhou et al. (2024) and aggregating the embedding similarities by taking the element-wise maximum instead of averaging them to ensure better recall.

3.2 Datasets

Our strategy is to re-use existing narrative classification dataset for the evaluation of narrative retrieval pipelines where we use narrative labels as queries and corresponding claims as gold labeled data. For our experiments, we chose these datasets based on the variety of topics and factors like language consistency, as the used datasets are in English, as well as data availability. The datasets we evaluate on are:

- The CARDS (Computer-assisted recognition of (climate change) denial and skepticism) dataset (Coan et al., 2021), comprised of climate change denial claims and neutral claims.
- The Climate Obstruction (hereafter referred to as CO) (Rowlands et al., 2024), comprised of climate obstruction in social media advertising samples.
- The PolyNarrative (hereafter referred to as PN) dataset (Nikolaidis et al., 2025), containing neutral news articles and articles containing Russia’s war on Ukraine propaganda and climate change denial.

While CARDS maps one narrative to each claim, CO and PN also account for multiple narratives being present in one text instance. Datasets like CARDS often follow predefined annotation codebooks with narrative taxonomies that are not necessarily selected by annotators. For example, in CARDS, 27 narratives were predefined in the codebook, while 17 were actually applied. Thus, for our experiments we only use narratives where corresponding texts can be found in the dataset. All three datasets are pre-split into training, development (dev) and testing sets. Except for PN where we use the dev set due to the absence of labels in the test set, we use the testing sets for our experiments.

3.3 Metrics

Retrieval Performance As documented in Table 1, all three datasets are imbalanced in terms of the amount R of texts mapped to each narrative. For example, in the CARDS dataset, R ranges between $R_{min} = 21$ and $R_{max} = 225$ with a standard deviation of 57.49 per narrative. When evaluating on the selected datasets, R is known and allows for the usage of metrics which incorporate R as a cutoff of retrieval metrics, namely R-Precision and $Precision@R_{min}$, $Recall@R_{min}$, $F1@R_{min}$. Each evaluation is performed over n narratives per dataset, which results in n individual R-Precision scores (e.g., in CARDS: 17 scores corresponding to 17 narratives). We aggregate these scores per run with:

- **Average R-Precision:** $\frac{1}{n}\sum_{i=1}^{n} \text{R-Precision}_i$, which weights all results per narrative equally.
- **Weighted R-Precision:** $\frac{\sum_{i=1}^{n} \text{TP}_i}{\sum_{i=1}^{n} R_i}$, which further accounts for class imbalance by giving more influence to narratives with more relevant documents.

Since we are specifically interested in moving towards real-life environments where R is not known, we also evaluate on more flexible metrics without reliance on R, namely on MAP and metrics with cutoffs set independent from R: nDCG@10, nDCG@100.

	n_k	mean R in N	std R in N	mean l in N	std l in N	mean l texts	std l texts	total texts	disinfo % of all
CARDS	17	67.65	57.49	7.61	3.96	65.35	57.60	2904	39.6
CO	7	38.29	27.20	20.50	3.56	28.27	11.95	255	73.3
PN	51	1.98	1.39	9.06	2.82	601.83	293.73	41	73.2
PN-UKR	27	2	1.47	9.32	3.30	740.62	382.17	13	100
PN-CC	23	2	1.32	8.77	2.12	495.71	120.74	17	100
PN-Neutral	0	-	-	-	-	459.78	166.97	11	0

Table 1: Statistics of the used datasets. N is the set of narratives with n_k elements. R is the amount of texts mapped to each n in N. l indicates the amount of detected words in each element

Complexity, Readability and Narrative Metrics Since our goal is to understand the notion of narratives better in the context of narrative retrieval, we analyze the datasets by different metrics collected in Table 2 to identify possible correlations between different factors. Inspired by the idea of measuring narrative distinctness in Irani et al. (2025), we also analyze the datasets with embedding-based metrics based on the calculation of different cosine distances:

Let $\mathcal{N} = \{n_1, \ldots, n_K\}$, $\mathbf{n}_{ij} \in \mathbb{R}^d$ be a set of narratives, represented by their sentence embeddings. Each narrative n_i has an associated set of text embeddings $\mathcal{T}_i = \{t_{i1}, \ldots, t_{im_i}\}$, $\mathbf{t}_{ij} \in \mathbb{R}^d$.

The centroid (mean embedding) of corresponding to each embedding n_i is

$$\mathbf{c}_i = \frac{1}{m_i}\sum_{j=1}^{m_i} \mathbf{t}_{ij} \in \mathbb{R}^d.$$

Cosine distance and cosine similarity for $\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$ are defined by

$$\text{cos\_dist}(\mathbf{a}, \mathbf{b}) = 1 - \text{cos\_sim}(\mathbf{a}, \mathbf{b}) = 1 - \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}$$

We measure **narrative distinctness** as proposed in (Irani et al., 2025):

$$\mathrm{D}_i(n_i) = \sqrt{\bar{d}_i \cdot \min d_i}$$

, where the inter-centroid distance is

$$d_{ij} = \text{cos\_dist}(\mathbf{c}_i, \mathbf{c}_j), \qquad j \neq i.$$

For narrative n_i, the mean and minimum distance to all other centroids are calculated with:

$$\bar{d}_i = \frac{1}{k-1}\sum_{j \neq i} D_{ij}, \qquad \min d_i = \min_{j \neq i} D_{ij}.$$

We propose the measurement of **narrative contingency** for measuring the space of contingent interpretations as the distance between a centroid $\mathbf{c}_i$ and the top-1 embedding from the set $\mathcal{T}_i$:

$$C_i = \text{cos\_dist}\left(\mathbf{c}_i, \mathbf{t}_i^{\text{top}}\right), \text{ where}$$

$$j_i^\star = \arg\max_{1 \leq j \leq m_i} \qquad \mathbf{t}_i^{\text{top}} = \mathbf{t}_{ij_i^\star}.$$

We measure **narrative variance** by calculating the variance between the text embeddings relative to their centroid:

$$\sigma^2 = \frac{1}{m_i}\sum_{j=1}^{m_i} \|\mathbf{t}_{ij} - \mathbf{c}_i\|_2^2.$$

We use gte-large (Li et al., 2023) for the creation of embeddings due to its relatively small size,

	smog	ari	mean len sentence	sentence complexity	mean D_i	mean C_i^1	mean σ^1
CARDS	13.84	14.27	16.64	0.53	0.04	0.06	0.17
CO	13.61	13.16	23.41	0.65	0.03	0.05	0.15
PN	14.14	14.09	28.97	0.68	0.05	0.02	0.05
PN-UKR	13.60	12.87	27.65	0.63	0.04	0.02	0.04
PN-CC	14.55	15.02	30.63	0.74	0.05	0.03	0.06
PN-Neutral	15.57	15.94	28.79	0.78	-	-	-

Table 2: Readability, complexity and embedding-based metrics on CARDS, CO and PolyNarrative datasets.

a larger embedding dimension of 1024 and no reliance on instruction prompts to minimize influence factors. The results are collected in Table 2.

Following studies such as Tavakoli et al., which indicate a possible correlation between results on readability metrics between texts with and without misinformation, we also collect SMOG and ARI following a recommendation from Stenger et al. (2023), which considered possible similarities between these metrics in specific cases. Additionally, we report the mean length of the sentences in each datasets in Table 2 and the mean sentence complexity, which we measured using a dependency parser (spaCy). A sentence was labeled complex if it contained at least one token with a subordinate/clausal dependency. The complex-sentence ratio equals complex sentences divided by total sentences.

3.4 Model choice

Within literature theory, the process of speculation is dependent on the reader, setting, and context. In our case, it is strongly influenced by the choice of LLM. Since proprietary models behind APIs are still state-of-the-art on many public leaderboards such as LMArena[1], we first explored options such as Gemini models and gpt-5 in our preliminary experiments, but faced one specific challenge of our proposed system: Many LLMs follow strict safety guidelines and thus refuse to generate hypothetical documents containing disinformation. We opted to work with the gpt-4omodel(OpenAI et al., 2024) because we evidenced less refusals during preliminary testing. Additionally, we wanted to include an evaluation based on an open source model to ensure reproducibility and to assess if the results differ between the original model and a publicly available "abliterated" version (a model version with safety alignment removed in post training). Here, we opted for the model gemma-3-27b-it (Team et al., 2025) due to its high ranking on LMArena. As with the original gemma model[2], we worked with a version quantized to 8 bits[3]. We chose the Qwen3-Embedding-4B model to generate embeddings in dense retrieval because of its good ranking on the MTEB[4] leaderboard, preliminary testing and the ability to include instruction prompts. In our setup which relied on proprietary models, we used OpenAI's text-embedding-3-large [5].

4 Evaluation

Retrieval Performance We first evaluate our system on performance metrics to further analyze possible correlations with other metrics introduced above. Table 3 documents our results, where averages of metrics over 10 runs are presented due to randomized factors within HyDE. In each evaluation, we treat the training sets as knowledge bases comprised of relevant data but without assessing the labels. For comparison, we also include the setups labeled with static* where few-shot examples were statically retrieved by assessing the labels.

Ablation Study To further analyze the influence of the components of our system, we run different ablation studies documented in Table 4. Here, our main goal is to provide comparison between the proposed SpecFi setups and results from sparse and dense retrieval only, since this performance metrics are also the base for our statistical analysis of correlation. We also include results for NodeRAG only, where we patched the framework to retrieve the full list of top k results directly.

Additional Datasets We further compare performance metrics and possible correlations with other metrics on two other datasets. The performance results are documented in Table 6 and Table 7. The PN dataset consists of two distinct topics and thus can be used to evaluate if the performance results differ between the topics. Selected metrics are documented in Table 8.

5 Statistical Analysis

As documented in Table 5, we can report several statistically significant correlations between performance on MAP and other metrics. The results

[1] https://lmarena.ai/leaderboard

[2] https://huggingface.co/unsloth/gemma-3-27b-it-GGUF

[3] https://huggingface.co/mlabonne/gemma-3-27b-it-abliterated-GGUF

[4] https://huggingface.co/spaces/mteb/leaderboard

[5] https://openai.com/index/new-embedding-models-and-api-updates/

Setup	#	Models	MAP	NDCG @10	NDCG @100	Weighted R-Precision	Average R-Precision	Precision @R$_{min}$	Recall @R$_{min}$	F1 @R$_{min}$	s / narrative
zero shot	1	4o, OI-E	0.321	0.509	0.487	0.343	0.370	0.433	0.181	0.255	29.091
	2	G,Q4B	0.313	0.469	0.456	0.332	0.371	0.399	0.167	0.236	18.867
	3	G-a,Q4B	0.295	0.428	0.436	0.304	0.308	0.362	0.159	0.221	17.205
static*	4	4o, OI-E	0.488	0.713	0.649	0.491	0.487	0.616	0.275	0.381	16.006
	5	G, Q4B	0.435	0.635	0.616	0.433	0.435	0.533	0.238	0.329	38.284
	6	G-a, Q4B	0.464	0.679	0.637	0.461	0.468	0.576	0.253	0.352	40.739
SpecFic -DR	7	4o, OI-E	0.421	0.682	0.600	0.432	0.440	0.552	0.239	0.333	17.19
	8	G, Q4B	0.424	0.630	0.581	0.429	0.432	0.555	0.247	0.342	39.321
	9	G-a, Q4B	0.457	0.693	0.619	0.457	0.453	0.584	0.263	0.363	34.902
SpecFic -NR	10	4o, OI-E	0.426	0.660	0.597	0.448	0.456	0.578	0.257	0.356	**15.124**
	11	G, Q4B	0.468	0.709	0.631	0.467	**0.492**	0.600	0.260	0.363	27.033
	12	G-a, Q4B	**0.505**	**0.738**	**0.660**	**0.495**	0.490	**0.638**	**0.287**	**0.395**	24.079

Table 3: Results on the CARDS dataset with $R_{min} = 21$. static* is included for reference only due to it's reliance on labels. F1 indicates the harmonic mean. All metrics are averaged over 10 runs. We report a standard deviation of <0.01 for all performance metrics. The model abbreviations are: 4o=gpt-4o, OI-E= text-embedding-3-large, G=gemma-3-27b-it, G-a=gemma-3-27b-it abliterated, Q4B=Qwen3-Embedding-4b.

vary per dataset, where narrative distinctness correlates negatively and significantly for SpecFi-NR-a on the CO dataset. On CARDS, we report significant negative correlation on narrative contingency and narrative variance, and a significant positive correlation of both on PN. On the other hand, no meaningful correlation can be reported in regard to SMOG and while there is a correlation between ARI and MAP on CARDS and CO, it is not statistically significant.

6 Discussion and Conclusions

In regard to **RQ1**, narrative classification datasets have shown to be viable options to be repurposed for the evaluation of retrieval tasks, providing human-annotated data with labeled relevant candidates to be retrieved by their key message. This allows for the measurement of different metrics, both dependent on the knowledge of R like R-Precision, and metrics like MAP which do not rely on this knowledge.

Regarding the general performance of our setups and **RQ2**, we notice BM25 yielding particularly low results on CARDS, as documented in Table 4. This includes the comparison against the dense retrieval only setups, which, as discussed above, is not necessarily a given. This could be attributed to the specific nature of this task, where the goal of interpretation has less utility for lexical matches. In most settings, our proposed speculation-based SpecFi setups outperform dense retrieval, highlighting the benefits of interpretative approaches in this setting. Surprisingly, community summaries are viable few-shot examples. As it can be seen in Table 3,in setups #11 and #12 this strategy actually outperformed the static sampling approach, which we only included for reference due to its reliance on labels. Here, using the community summaries yielded better results than actually labeled examples from the training split of the same dataset. There are, however, differences across datasets: As with BM25, the performance gaps are larger on CARDS and smaller on CO and PN, which could be explained by both, the pool of potential candidates and the pool of neutral candidates being substantially smaller in CO and PN. When working with smaller datasets, dense retrieval could still be a viable option, especially when considering the high runtime requirements of the speculation-based approaches.

Setup/Model	MAP	NDCG @10	Average R-Precision	Weighted R-Precision	F1@ R$_{min}$	s / narrative
NodeRAG only	0.259	0.506	0.323	0.290	0.222	1.931
BM25	0.080	0.125	0.119	0.106	0.064	0.011
thenlper/gte-large	0.215	0.394	0.272	0.236	0.171	2.092
OpenAI-E	0.262	0.507	0.323	0.287	0.203	0.452
Qwen3-E-4B	0.299	0.523	0.352	0.321	0.238	6.645
Qwen3-E-4B-p	0.316	0.536	0.370	0.343	0.263	6.593

Table 4: Ablation. Models: OpenAI-E=text-embedding-3-large, Qwen3-E-4B=Qwen3-Embedding-4b

Our statistical analysis in regard to **RQ3** indicates correlations between how each narrative group is constructed and the performance results. Generally, MAP results can get worse with higher narrative contingency and narrative variance results on CARDS, while having a positive influence on

		Narrative Distinctness			Narrative Contingency			Narrative Variance			SMOG			ARI		
Metric	Setup	CARDS	CO	PN	CARDS	CO	PN	CARDS	CO	PN	CARDS	CO	PN	CARDS	CO	PN
Spearman's rho	BM25	-0.240	-0.357	0.369	-0.520	-0.143	0.321	-0.525	-0.071	0.319	0.115	0.214	-0.019	0.449	0.750	-0.019
	QWEN-E-4B	-0.066	-0.679	0.197	-0.549	-0.071	0.122	-0.556	0.000	0.151	-0.020	0.571	-0.159	0.412	0.857	-0.188
	SpecFi-DR-a	0.147	-0.964	-0.016	-0.603	0.250	0.373	-0.578	0.214	0.476	-0.110	0.179	0.044	0.221	0.179	0.042
	SpecFi-NR-a	0.282	-0.786	-0.041	-0.370	-0.214	0.170	-0.324	-0.286	0.249	-0.194	-0.107	0.024	0.157	-0.143	0.024
p-value	BM25	0.441	0.536	**0.014**	**0.041**	0.879	**0.027**	**0.041**	0.977	**0.028**	0.748	0.877	0.896	0.118	0.130	0.894
	QWEN-E-4B	0.801	0.188	0.208	**0.040**	0.879	0.394	**0.040**	1.000	0.291	0.940	0.538	0.378	0.144	0.130	0.265
	SpecFi-DR-a	0.637	0.121	0.913	**0.026**	0.879	**0.014**	**0.037**	0.891	**0.001**	0.748	0.879	0.896	0.439	0.844	0.894
	SpecFi-NR-a	0.390	**0.005**	0.863	0.149	0.879	0.258	0.205	0.921	0.087	0.652	0.877	0.896	0.548	0.844	0.894

Table 5: Statistical analysis results in regard to the correlation between performance on MAP, embedding-based and readability metrics. The results document FDR-corrected p-values. Bold values refer to statistical significance where p<0.05.

PN. This could be related to the amount of texts which are mapped to each narrative per dataset, documented in Table 1: In PN, the mean amount of texts mapped to each narrative is only 1.98, while the total amount of labeled examples is 41, out of which 30 have 51 disinformation narratives labeled to them. Compared to CARDS with 2904 texts mapped to 17 narratives and a mean of 67.65 potential candidates. Thus, while our findings indicate that texts can in fact be retrieved by their key messages, these need to be sufficiently distinct. In future work, this evaluation strategy can be used for the refinement of narrative taxonomies, pointing out directions of where narrative groups with large variance could be split in two, for example.

While the correlation of performance and readability metrics is not statistically significant, a relation can be interpreted based on the results where the metrics are compared by each topic in the PN dataset. All proposed setups in Table 8 perform better on War in Ukraine propaganda. This is especially noticeable with the performance gap of 0.161 points between topics dense retrieval only. In Table 2, we can report a lower ARI score for this topic by over 3 points, meaning that the climate change denial texts are harder to read. The corresponding performance drop aligns with the higher Spearman's rho value. This could imply that War in Ukraine disinformation texts are less complex, are more superficial and thus easier to retrieve, actually enabling simple dense retrieval strategies to outperform more advanced pipelines.

Table 3 showcases our results on CARDS in regard to **RQ4**. The setup #12 featured an abliterated model and outperformed all other configurations across most metrics, In most cases where the abliterated model was deployed, the best performance metrics were yielded. However, the results on other datasets are more ambiguous. On the PN dataset (Table 7), SpecFi-NR-a outperformed the setup with the original gemma model, but the same could not be observed with the SpecFi-DR setups. On CO, documented in Table 6, all results are on a similar level, while the un-abliterated settings slightly outperform on some metrics.

Setup	MAP	NDCG @10	Average R-Precision	Weighted R-Precision	F1 @R_{min}	s / narrative
BM25	0.311	0.378	0.219	0.248	0.129	**0.007**
Qwen3-E-4B	**0.502**	0.598	0.374	0.436	0.314	0.233
SpecFi-DR	0.443	0.621	0.370	0.404	0.317	400.752
SpecFi-DR-a	0.386	0.536	0.275	0.362	0.216	330.915
SpecFi-NR	0.458	0.626	0.372	**0.443**	0.334	17.550
SpecFi-NR-a	0.471	**0.640**	**0.386**	0.417	**0.342**	66.383

Table 7: Evaluation on PolyNarrative dataset with $R_{min} = 1$

Setup	MAP	NDCG @10	Average R-Precision	Weighted R-Precision	F1 @R_{min}	s / narrative
bm25	0.326	0.472	0.298	0.362	0.111	**0.001**
Qwen3-E-4B	0.499	0.607	0.491	**0.604**	0.088	0.113
SpecFi-DR	**0.519**	**0.644**	**0.496**	0.565	**0.209**	18.24
SpecFi-DR-a	0.482	0.604	0.494	0.567	0.197	17.217
SpecFi-NR	0.491	0.618	0.49	0.566	0.177	20.766
SpecFi-NR-a	0.495	0.627	0.486	0.567	0.249	19.424

Table 6: Evaluation on Climate Obstruction dataset with $R_{min} = 3$. Qwen3-E-4B=Qwen3-Embedding-4B

Topic	Amount	Setup	MAP	Average R-Precision	Weighted R-Precision
War in Ukraine	28/51	BM25	0.333	0.235	0.290
		Qwen3-E-4B	**0.590**	**0.464**	**0.509**
		SpecFic-NR	0.429	0.410	0.473
		SpecFic-NR-a	0.447	0.415	0.447
Climate Change	23/51	BM25	0.283	0.199	0.197
		Qwen3-E-4B	0.445	0.297	0.348
		SpecFic-NR	0.492	0.336	**0.409**
		SpecFic-NR-a	**0.500**	**0.355**	0.396

Table 8: PN by topic. Bold values indicate the best results per topic

Limitations

As shown in Table 1, the used datasets are similar in nature, but differ vastly in factors such as the narrative-to-text ratios. This aligns with our results, where different performance and correlations with other metrics were measured, hinting at a potential research gap on how to harmonize such datasets. Measures like narrative distinctness and the proposed metrics narrative contingency and narrative variance can be applied to revise taxonomies during harmonization efforts.

While we were able to provide a version of SpecFi-DR which only relies on open source models to ensure reproducibility, the SpecFi-NR setups include one reliance on OpenAI models within NodeRAG. Recent studies have shown that OpenAI models still outperform on structured output generation (Geng et al., 2025), which is an essential step during graph construction. For this reason and due to NodeRAG's own recommendation[6], we used the recommended proprietary model here.

Another influence could limit our performance results: the CARDS dataset is from 2021, making it likely to be part of the training data of LLMs. While this does not necessarily relate to our specific usage of this dataset, it is still possible that there is an influence on the generation of hypothetical documents. However, our results of the zero shot variants in Table 3 indicate that no of our tested LLMs is capable of generating representative hypotheticals without examples and only based on the narrative by itself, but an influence of some kind of capacity cannot be ruled out.

Although HyDE-based techniques lead to better performance, the final retrieval step still relies on dense retrieval. Following (Akter and Santu, 2024), concrete narrative similarity scores beyond cosine similarity and towards more interpretative retrieval could be explored further, which also incorporate the extraction and assessment of more narrative features, like actors, localities and other story features. Building up on this, while adding graph reasoning steps did improved performance, the applied approach was an extracted feature from NodeRAG, which was used in an out-of-the-box manner. Representing the graph data more aligned with narrative features representing narrative systems as discussed in (Hellman, 2024) could further improve the performance and understanding of this narrative context.

[6] https://terry-xu-666.github.io/NodeRAG_web/blog/2025/03/16/structure-output/

Ethical Considerations

The narrative retrieval approach proposed in this paper is dependent on LLMs that do not refuse to generate disinformation. Although our goal is to identify disinformation to combat it, our method could also encourage enhanced fine-tuning of LLMs to improve generating disinformation. Within this study, we only use models already available in huggingface. But this point needs to be taken into account further in future work, like the question of whether models fine-tuned for generating disinformation should be released publicly.

References

Mousumi Akter and Shubhra Kanti Karmaker Santu. 2024. Fans: a facet-based narrative similarity metric. *Preprint*, arXiv:2309.04823.

Boyu Chen, Zirui Guo, Zidan Yang, Yuluo Chen, Junze Chen, Zhenghao Liu, Chuan Shi, and Cheng Yang. 2025. Pathrag: Pruning graph-based retrieval augmented generation with relational paths. *Preprint*, arXiv:2502.14902.

Travis G. Coan, Constantine Boussalis, John Cook, and Mirjam O. Nanko. 2021. Computer-assisted classification of contrarian claims about climate change. *Scientific Reports*, 11(1):22320.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitansky, Robert Osazuwa Ness, and Jonathan Larson. 2025. From local to global: A graph rag approach to query-focused summarization. *Preprint*, arXiv:2404.16130.

Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. 2023. Precise zero-shot dense retrieval without relevance labels. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1762–1777, Toronto, Canada. Association for Computational Linguistics.

Yao Ge, Sudeshna Das, Yuting Guo, and Abeed Sarker. 2025. Retrieval augmented generation based dynamic prompting for few-shot biomedical named entity recognition using large language models. *Preprint*, arXiv:2508.06504.

Saibo Geng, Hudson Cooper, Michał Moskal, Samuel Jenkins, Julian Berman, Nathan Ranchin, Robert West, Eric Horvitz, and Harsha Nori. 2025. Jsonschemabench: A rigorous benchmark of structured outputs for language models. *Preprint*, arXiv:2501.10868.

Zirui Guo, Lianghao Xia, Yanhua Yu, Tu Ao, and Chao Huang. 2025. Lightrag: Simple and fast retrieval-augmented generation. *Preprint*, arXiv:2410.05779.

Shailja Gupta, Rajesh Ranjan, and Surya Narayan Singh. 2024. A comprehensive survey of retrieval-augmented generation (rag): Evolution, current landscape and future directions. *Preprint*, arXiv:2410.12837.

Fatima Haouari, Carolina Scarton, Nicolò Faggiani, Nikolaos Nikolaidis, Bonka Kotseva, Ibrahim Abu Farha, Jens Linge, and Kalina Bontcheva. 2025. UK-ElectionNarratives: A Dataset of Misleading Narratives Surrounding Recent UK General Elections. *Proceedings of the International AAAI Conference on Web and Social Media*, 19:2477–2495.

Philipp Heinrich, Andreas Blombach, Bao Minh Doan Dang, Leonardo Zilio, Linda Havenstein, Nathan Dykes, Stephanie Evert, and Fabian Schäfer. 2024. Automatic identification of COVID-19-related conspiracy narratives in German telegram channels and chats. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 1932–1943, Torino, Italia. ELRA and ICCL.

Maria Hellman. 2024. *Narrative Analysis and Framing Analysis of Disinformation*, pages 101–121. Springer Nature Switzerland, Cham.

Pavan Holur, Tianyi Wang, Shadi Shahsavari, Timothy Tangherlini, and Vwani Roychowdhury. 2022. Which side are you on? insider-outsider classification in conspiracy-theoretic social media. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4975–4987, Dublin, Ireland. Association for Computational Linguistics.

Arman Irani, Ju Yeon Park, Kevin Esterling, and Michalis Faloutsos. 2025. A discourse analysis framework for legislative and social media debates. In *Proceedings of the 17th ACM Web Science Conference 2025*, Websci '25, page 199–209, New York, NY, USA. Association for Computing Machinery.

Cody Kommers, Ruth Ahnert, Maria Antoniak, Emmanouil Benetos, Steve Benford, Mercedes Bunz, Baptiste Caramiaux, Shauna Concannon, Martin Disley, James Dobson, Yali Du, Edgar Duéñez-Guzmán, Kerry Francksen, Evelyn Gius, Jonathan Gray, Ryan Heuser, Sarah Immel, Richard So, Sang Leigh, and 19 others. 2025. Computational Hermeneutics: Evaluating Generative AI as a Cultural Technology.

Bonka Kotseva, Irene Vianini, Nikolaos Nikolaidis, Nicolò Faggiani, Kristina Potapova, Caroline Gasparro, Yaniv Steiner, Jessica Scornavacche, Guillaume Jacquet, Vlad Dragu, Leonida della Rocca, Stefano Bucci, Aldo Podavini, Marco Verile, Charles Macmillan, and Jens P. Linge. 2023. Trend analysis of COVID-19 mis/disinformation narratives–A 3-year study. *PLOS ONE*, 18(11):e0291423. Publisher: Public Library of Science.

Yuanyuan Lei and Ruihong Huang. 2023. Discourse structures guided fine-grained propaganda identification. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 331–342, Singapore. Association for Computational Linguistics.

Siran Li, Linus Stenzel, Carsten Eickhoff, and Seyed Ali Bahrainian. 2025a. Enhancing retrieval-augmented generation: A study of best practices. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 6705–6717, Abu Dhabi, UAE. Association for Computational Linguistics.

Yangning Li, Weizhi Zhang, Yuyao Yang, Wei-Chieh Huang, Yaozu Wu, Junyu Luo, Yuanchen Bei, Henry Peng Zou, Xiao Luo, Yusheng Zhao, Chunkit Chan, Yankai Chen, Zhongfen Deng, Yinghui Li, Hai-Tao Zheng, Dongyuan Li, Renhe Jiang, Ming Zhang, Yangqiu Song, and Philip S. Yu. 2025b. Towards agentic rag with deep reasoning: A survey of rag-reasoning systems in llms. *Preprint*, arXiv:2507.09477.

Zehan Li, Xin Zhang, Yanzhao Zhang, Dingkun Long, Pengjun Xie, and Meishan Zhang. 2023. Towards general text embeddings with multi-stage contrastive learning. *Preprint*, arXiv:2308.03281.

Nikolaos Nikolaidis, Nicolas Stefanovitch, Purificação Silvano, Dimitar Iliyanov Dimitrov, Roman Yangarber, Nuno Guimarães, Elisa Sartori, Ion Androutsopoulos, Preslav Nakov, Giovanni Da San Martino, and Jakub Piskorski. 2025. PolyNarrative: A multilingual, multilabel, multi-domain dataset for narrative extraction from news articles. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 31323–31345, Vienna, Austria. Association for Computational Linguistics.

OpenAI, :, Aaron Hurst, Adam Lerer, Adam P. Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, Aleksander Mądry, Alex Baker-Whitcomb, Alex Beutel, Alex Borzunov, Alex Carney, Alex Chow, Alex Kirillov, and 401 others. 2024. Gpt-4o system card. *Preprint*, arXiv:2410.21276.

Andrew Piper, Richard Jean So, and David Bamman. 2021. Narrative theory for computational narrative understanding. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 298–311, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Ruiyang Ren, Yingqi Qu, Jing Liu, Xin Zhao, Qifei Wu, Yuchen Ding, Hua Wu, Haifeng Wang, and Ji-Rong Wen. 2023. A thorough examination on zero-shot dense retrieval. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages

15783–15796, Singapore. Association for Computational Linguistics.

Hanna-Riikka Roine. 2020. On speculation as a strategy. *Fafnir – Nordic Journal of Science Fiction and Fantasy Research*, 7(2):8–15.

Harri Rowlands, Gaku Morio, Dylan Tanner, and Christopher Manning. 2024. Predicting narratives of climate obstruction in social media advertising. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 5547–5558, Bangkok, Thailand. Association for Computational Linguistics.

Yanzhen Shen, Sihao Chen, Xueqiang Xu, Yunyi Zhang, Chaitanya Malaviya, and Dan Roth. 2025. Logicol: Logically-informed contrastive learning for set-based dense retrieval. *Preprint*, arXiv:2505.19588.

Iknoor Singh. 2024. *Detecting and Tracking the Spread of Debunked Narratives Across Languages*. Phd thesis, University of Sheffield, Sheffield, UK. Supervisors: Carolina Scarton and Kalina Bontcheva.

Iknoor Singh, Carolina Scarton, Xingyi Song, and Kalina Bontcheva. 2024. Breaking language barriers with mmtweets: Advancing cross-lingual debunked narrative retrieval for fact-checking. *Preprint*, arXiv:2308.05680.

Veronika Solopova, Christoph Benzmüller, and Tim Landgraf. 2023a. The evolution of pro-kremlin propaganda from a machine learning and linguistics perspective. In *Proceedings of the Second Ukrainian Natural Language Processing Workshop (UNLP)*, pages 40–48, Dubrovnik, Croatia. Association for Computational Linguistics.

Veronika Solopova, Oana-Iuliana Popescu, Christoph Benzmüller, and Tim Landgraf. 2023b. Automated multilingual detection of pro-kremlin propaganda in newspapers and telegram posts. *Datenbank-Spektrum*, 23(1):5–14.

Witold Sosnowski, Arkadiusz Modzelewski, Kinga Skorupska, Jahna Otterbacher, and Adam Wierzbicki. 2024. EU DisinfoTest: a benchmark for evaluating language models' ability to detect disinformation narratives. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 14702–14723, Miami, Florida, USA. Association for Computational Linguistics.

Rachel Stenger, Kristen Olson, and Jolene D. Smyth. 2023. Comparing readability measures and computer-assisted question evaluation tools for self-administered survey questions. *Field Methods*, 35(4):287–302.

Hongjin Su, Howard Yen, Mengzhou Xia, Weijia Shi, Niklas Muennighoff, Han yu Wang, Haisu Liu, Quan Shi, Zachary S. Siegel, Michael Tang, Ruoxi Sun, Jinsung Yoon, Sercan O. Arik, Danqi Chen, and Tao Yu. 2025. Bright: A realistic and challenging benchmark for reasoning-intensive retrieval. *Preprint*, arXiv:2407.12883.

Melanie Subbiah, Sean Zhang, Lydia B. Chilton, and Kathleen McKeown. 2024. Reading subtext: Evaluating large language models on short story summarization with writers. *Transactions of the Association for Computational Linguistics*, 12:1290–1310.

Mohammadali Tavakoli, Harith Alani, and Grégoire Burel. On the Readability of Misinformation in Comparison to the Truth.

Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, Louis Rouillard, Thomas Mesnard, Geoffrey Cideron, Jean bastien Grill, Sabela Ramos, Edouard Yvinec, Michelle Casbon, Etienne Pot, Ivo Penchev, and 197 others. 2025. Gemma 3 technical report. *Preprint*, arXiv:2503.19786.

Nandan Thakur, Luiz Bonifacio, Maik Fröbe, Alexander Bondarenko, Ehsan Kamalloo, Martin Potthast, Matthias Hagen, and Jimmy Lin. 2024. Systematic evaluation of neural retrieval models on the touché 2020 argument retrieval subset of beir. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '24, page 1420–1430, New York, NY, USA. Association for Computing Machinery.

Orion Weller, Michael Boratko, Iftekhar Naim, and Jinhyuk Lee. 2025. On the theoretical limitations of embedding-based retrieval. *Preprint*, arXiv:2508.21038.

Tianyang Xu, Haojie Zheng, Chengze Li, Haoxiang Chen, Yixin Liu, Ruoxi Chen, and Lichao Sun. 2025. Noderag: Structuring graph-based rag with heterogeneous nodes. *Preprint*, arXiv:2504.11544.

Hao Yu, Aoran Gan, Kai Zhang, Shiwei Tong, Qi Liu, and Zhaofeng Liu. 2025. Evaluation of retrieval-augmented generation: A survey. In *Big Data*, pages 102–120, Singapore. Springer Nature Singapore.

Yan Zhang, Zhaopeng Feng, Zhiyang Teng, Zuozhu Liu, and Haizhou Li. 2023. How well do text embedding models understand syntax? In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 9717–9728, Singapore. Association for Computational Linguistics.

Weichao Zhou, Jiaxin Zhang, Hilaf Hasson, Anu Singh, and Wenchao Li. 2024. Hyqe: Ranking contexts with hypothetical query embeddings. *Preprint*, arXiv:2410.15262.

A Appendix

A.1 Evaluation Details

A.1.1 Additional Statistical Analysis of Performance Correlation

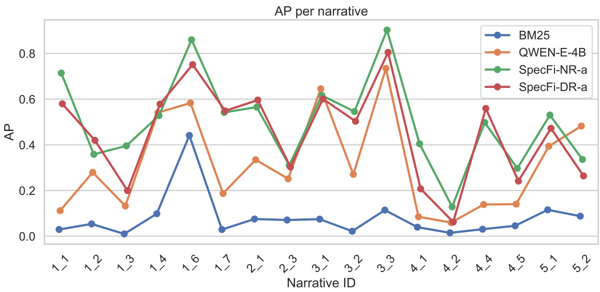

Figure 2: AP results per narrative on the CARDS dataset

As documented in Figure 2, although the retrieval performance differs per narrative when compared over the whole CARDS dataset, there are also similarities between the results across setups. For example, the two best results for all setups are the spikes at the narrative ids 1_6 and 3_3, while 4_2 is one of the lowest scores for all 4 setups. In Table 9, we evaluate if the performance of the considered systems does indeed correlate. Several statistically significant correlations can be reported, especially within the results on CARDS and PN. For example, the comparison between BM25 QWEN3-E-4B indicates the highest correlation with a rho value of 0.824 and a FDR-corrected p-value of 0.000 on CARDS. Both SpecFi variants behave more independently, especially in regard to the SpecFI-NR-a setup which, for example, yields the lowest rho values when compared to BM25 results with a p-value of 0.249 on CARDS and thus with the highest value above the 0.05 significance threshold.

#	Metric	Setup	BM25	QWEN -E-4B	SpecFi -NR-a	SpecFi -DR-a
1	RHO	BM25	1.000	0.824	0.365	0.544
		QWEN-E-4B	0.824	1.000	0.520	0.679
		SpecFi-NR-a	0.365	0.520	1.000	0.892
		SpecFi-DR-a	0.544	0.679	0.892	1.000
	FDR-p	BM25	0.000	**0.000**	0.249	0.060
		QWEN-E-4B	**0.000**	0.000	0.065	**0.009**
		SpecFi-NR-a	0.249	0.065	0.000	**0.000**
		SpecFi-DR-a	0.060	**0.009**	**0.000**	0.000
2	RHO	BM25	1.000	0.786	-0.071	0.321
		QWEN-E-4B	0.786	1.000	0.321	0.536
		SpecFi-NR-a	-0.071	0.321	1.000	0.750
		SpecFi-DR-a	0.321	0.536	0.750	1.000
	FDR-p	BM25	0.000	0.121	0.879	0.536
		QWEN-E-4B	0.121	0.000	0.536	0.359
		SpecFi-NR-a	0.879	0.536	0.000	0.130
		SpecFi-DR-a	0.536	0.359	0.130	0.000
3	RHO	BM25	1.000	0.657	0.366	0.489
		QWEN-E-4B	0.657	1.000	0.353	0.436
		SpecFi-NR-a	0.366	0.353	1.000	0.558
		SpecFi-DR-a	0.489	0.436	0.558	1.000
	FDR-p	BM25	0.000	**0.000**	**0.014**	**0.001**
		QWEN-E-4B	**0.000**	0.000	**0.016**	**0.004**
		SpecFi-NR-a	**0.014**	**0.016**	0.000	**0.000**
		SpecFi-DR-a	**0.001**	**0.004**	**0.000**	0.000

Table 9: Spearman's rho and FDR-corrected p-values for correlation analysis of performances on the MAP metric. #1 documents the results on CARDS, #2 on CO and #3 on PN.

A.2 Prompts Collection

A.2.1 Embedding models

The default model instruction prompt is:

```
f``Instruct: Given a web search query, retrieve relevant passages that answer the query \n
Query: {query}''
```

The following prompt was used to enhance the retrieval results:

```
f``Instruct: Given a narrative description as a query, retrieve passages that serve this narrative;
can be entailed from the narrative;can be aligned logically with the narrative \n Query: {query}''
```

Within our HyDE-based setup, we use the following instruction prompt for embedding models:

```
f``Instruct: Given a text as a query retrieve relevant passages that align with narratives
similar to the query\n Query: {query}''
```

A.2.2 Generative Models

Our system prompt:

```
f``You are a disinformation investigator. Your first step is to generate short disinformation texts
that sound like actual ones. You get a disinformation narrative and return a disinformation text
that aligns with that narrative. Return only 1 single text!''
```

Each system prompt was followed by a user prompt:

```
f``You are a disinformation investigator. Given a disinformation narrative, generate a short,
realistic text (such as a news excerpt, blog post, or social media post) that supports or aligns
with that narrative. The text should sound plausible and could be found in the wild.
\n\n Here are some examples: \{examples\}
\n\nNarrative: \{query\} \nText: ''
```

A.3 Narrative Taxonomies transformed to Search Queries

All datasets included predefined narrative taxonomies. We constructed narrative descriptions from the narrative labels with the following strategies:

A.3.1 CARDS

The queries were constructed by following the CARDS taxonomy and the corresponding id labeling with the dataset and concatenating "claim" (=subnarrative) and "superclaim"(=narrative) descriptions from the code book.

```
"1_1": "Global warming is not happening. Ice/permafrost/snow cover isn't melting",
"1_2": "Global warming is not happening. We're heading into an ice age/global cooling"
"1_3": "Global warming is not happening. Weather is cold/snowing",
...
"5_3": "Climate movement/science is unreliable. Climate change (science or policy) is a conspiracy
        (deception)",
```

Resulting in predefined 27 narratives in the taxonomy, out of which 17 can be found in the dataset.

A.3.2 Climate Obstruction

Constructed by using the narrative ids from the dataset and descriptions of the narratives provided in the supplemental material of the paper.

```
"CA": "Community & Resilience. Emphasizes how the oil and gas sector contributes to local and national
      economies through tax revenues, charitable efforts, and support for local businesses",
"CB": "Community & Resilience. Focuses on the creation and sustainability of jobs by the oil
      and gas industry."
...
"SA": "Patriotic Energy mix. Stresses how domestic oil and gas production benefits the nation,
      including energy independence, energy leadership, and the idea
      of supporting American energy"
```

Resulting in 6 narratives, out of which all can be found in the dataset.

A.3.3 PolyNarrative

Similar to CARDS, the queries were constructed by following the PolyNarrative (PN taxonomy and the corresponding id labeling with the dataset.

```
"1_1": "Blaming the war on others rather than the invader: Ukraine is the aggressor",
"1_2": "Blaming the war on others rather than the invader: The West are the aggressors",
...
"21_2": "Green policies are geopolitical instruments: Green activities are a form of neo-colonialism"
```

Resulting in predefined 88 narratives in the taxonomy, out of which 51 can be found in the dataset.